PART IX

Factorization

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SECTION 45 UNIQUE FACTORIZATION DOMAINS

The integral domain $\mathbb{Z}$ is our standard example of an integral domain in which there is unique factorization into primes (irreducibles). Section 23 showed that for a field F, $F[x]$ is also such an integral domain with unique factorization. In order to discuss analogous ideas in an arbitrary integral domain, we shall give several definitions, some of which are repetitions of earlier ones. It is nice to have them all in one place for reference.

45.1 Definition Let R be a commutative ring with unity and let $a, b \in R$. If there exists $c \in R$ such that $b = ac$, then a **divides** b (or a **is a factor of** b), denoted by $a \mid b$. We read $a \nmid b$ as "a does not divide b." ■

45.2 Definition An element u of a commutative ring with unity R is a **unit of** R if u divides 1, that is, if u has a multiplicative inverse in R. Two elements $a, b \in R$ are **associates in** R if $a = bu$, where u is a unit in R.

Exercise 27 asks us to show that this criterion for a and b to be associates is an equivalence relation on R. ■

45.3 Example The only units in $\mathbb{Z}$ are 1 and -1. Thus the only associates of 26 in $\mathbb{Z}$ are 26 and -26. ▲

45.4 Definition A nonzero element p that is not a unit of an integral domain D is an **irreducible of** D if in every factorization $p = ab$ in D has the property that either a or b is a unit. ■

Note that an associate of an irreducible p is again an irreducible, for if $p = uc$ for a unit u, then any factorization of c provides a factorization of p.

■ Historical Note

The question of unique factorization in an integral domain other than the integers was first raised in public in connection with the attempted proof by Gabriel Lamé (1795–1870) of Fermat's Last Theorem, the conjecture that $x^n + y^n = z^n$ has no nontrivial integral solutions for $n > 2$. It is not hard to show that the conjecture is true if it can be proved for all odd primes p. At a meeting of the Paris Academy on March 1, 1847, Lamé announced that he had proved the theorem and presented a sketch of the proof. Lamé's idea was first to factor $x^p + y^p$ over the complex numbers as

$$x^p + y^p = (x + y)(x + \alpha y)(x + \alpha^2 y) \cdots (x + \alpha^{p-1} y)$$

where α is a primitive pth root of unity. He next proposed to show that if the factors in this expression are relatively prime and if $x^p + y^p = z^p$, then each of the p factors must be a pth power. He could then demonstrate that this Fermat equation would be true for a triple x', y', z', each number smaller than the corresponding number in the original triple. This would lead to an infinite descending sequence of positive integers, an impossibility that would prove the theorem.

After Lamé finished his announcement, however, Joseph Liouville (1809–1882) cast serious doubts on the purported proof, noting that the conclusion that each of the relatively prime factors was a pth power because their product was a pth power depended on the result that any integer can be uniquely factored into a product of primes. It was by no means clear that "integers" of the form $x + \alpha^k y$ had this unique factorization property. Although Lamé attempted to overcome Liouville's objections, the matter was settled on May 24, when Liouville produced a letter from Ernst Kummer noting that in 1844 he had already proved that unique factorization failed in the domain $\mathbb{Z}[\alpha]$, where α is a 23rd root of unity.

It was not until 1994 that Fermat's Last Theorem was proved, and by techniques of algebraic geometry unknown to Lamé and Kummer. In the late 1950s, Yutaka Taniyama and Goro Shimura noticed a curious relationship between two seemingly disparate fields of mathematics, elliptic curves and modular forms. A few years after Taniyama's tragic death at age 31, Shimura clarified this idea and eventually formulated what became known as the Taniyama–Shimura Conjecture. In 1984, Gerhard Frey asserted and in 1986 Ken Ribet proved that the Taniyama–Shimura Conjecture would imply the truth of Fermat's Last Theorem. But it was finally Andrew Wiles of Princeton University who, after secretly working on this problem for seven years, gave a series of lectures at Cambridge University in June 1993 in which he announced a proof of enough of the Taniyama–Shimura Conjecture to derive Fermat's Last Theorem. Unfortunately, a gap in the proof was soon discovered, and Wiles went back to work. It took him more than a year, but with the assistance of his student Richard Taylor, he finally was able to fill the gap. The result was published in the *Annals of Mathematics* in May 1995, and this 350-year-old problem was now solved.

45.5 Definition An integral domain D is a **unique factorization domain** (abbreviated UFD) if the following conditions are satisfied:

1. Every element of D that is neither 0 nor a unit can be factored into a product of a finite number of irreducibles.
2. If $p_1 \cdots p_r$ and $q_1 \cdots q_s$ are two factorizations of the same element of D into irreducibles, then $r = s$ and the q_j can be renumbered so that p_i and q_i are associates.

■

45.6 Example Theorem 23.20 shows that for a field F, $F[x]$ is a UFD. Also we know that $\mathbb{Z}$ is a UFD: we have made frequent use of this fact, although we have never proved it. For example, in $\mathbb{Z}$ we have

$$24 = (2)(2)(3)(2) = (-2)(-3)(2)(2).$$

Here 2 and -2 are associates, as are 3 and -3. Thus except for order and associates, the irreducible factors in these two factorizations of 24 are the same. ▲

Recall that the *principal ideal* $\langle a \rangle$ of D consists of all multiples of the element a. After just one more definition we can describe what we wish to achieve in this section.

45.7 Definition An integral domain D is a **principal ideal domain** (abbreviated PID) if every ideal in D is a principal ideal. ■

We know that $\mathbb{Z}$ is a PID because every ideal is of the form $n\mathbb{Z}$, generated by some integer n. Theorem 27.24 shows that if F is a field, then $F[x]$ is a PID. Our purpose in this section is to prove two exceedingly important theorems:

1. Every PID is a UFD. (Theorem 45.17)
2. If D is a UFD, then $D[x]$ is a UFD. (Theorem 45.29)

The fact that $F[x]$ is a UFD, where F is a field (by Theorem 23.20), illustrates both theorems. For by Theorem 27.24, $F[x]$ is a PID. Also, since F has no nonzero elements that are not units, F satisfies our definition for a UFD. Thus Theorem 45.29 would give another proof that $F[x]$ is a UFD, except for the fact that we shall actually use Theorem 23.20 in proving Theorem 45.29. In the following section we shall study properties of a certain special class of UFDs, the *Euclidean domains*.

Let us proceed to prove the two theorems.

Every PID Is a UFD

The steps leading up to Theorem 23.20 and its proof indicate the way for our proof of Theorem 45.17. Much of the material will be repetitive. We inefficiently handled the special case of $F[x]$ separately in Theorem 23.20, since it was easy and was the only case we needed for our field theory in general.

To prove that an integral domain D is a UFD, it is necessary to show that both Conditions 1 and 2 of the definition of a UFD are satisfied. For our special case of $F[x]$ in Theorem 23.20, Condition 1 was very easy and resulted from an argument that in a factorization of a polynomial of degree > 0 into a product of two nonconstant polynomials, the degree of each factor was less than the degree of the original polynomial. Thus we couldn't keep on factoring indefinitely without running into unit factors, that is, polynomials of degree 0. For the general case of a PID, it is harder to show that this is so. We now turn to this problem. We shall need one more set-theoretic concept.

45.8 Definition If $\{A_i \mid i \in I\}$ is a collection of sets, then the **union** $\cup_{i \in I} A_i$ **of the sets** A_i is the set of all x such that $x \in A_i$ for at least one $i \in I$. ■

45.9 Lemma Let R be a commutative ring and let $N_1 \subseteq N_2 \subseteq \cdots$ be an ascending chain of ideals N_i in R. Then $N = \cup_i N_i$ is an ideal of R.

Proof Let $a, b \in N$. Then there are ideals N_i and N_j in the chain, with $a \in N_i$ and $b \in N_j$. Now either $N_i \subseteq N_j$ or $N_j \subseteq N_i$; let us assume that $N_i \subseteq N_j$, so both a and b are in N_j. This implies that $a \pm b$ and ab are in N_j, so $a \pm b$ and ab are in N. Taking $a = 0$, we see that $b \in N$ implies $-b \in N$, and $0 \in N$ since $0 \in N_i$. Thus N is a subring of D. For $a \in N$ and $d \in D$, we must have $a \in N_i$ for some N_i. Then since N_i is an ideal, $da = ad$ is in N_i. Therefore, $da \in \cup_i N_i$, that is, $da \in N$. Hence N is an ideal. ◆

45.10 Lemma **(Ascending Chain Condition for a PID)** Let D be a PID. If $N_1 \subseteq N_2 \subseteq \cdots$ is an ascending chain of ideals N_i, then there exists a positive integer r such that $N_r = N_s$ for all $s \geq r$. Equivalently, every strictly ascending chain of ideals (all inclusions proper) in a PID is of finite length. We express this by saying that the **ascending chain condition** (ACC) holds for ideals in a PID.

Proof By Lemma 45.9, we know that $N = \cup_i N_i$ is an ideal of D. Now as an ideal in D, which is a PID, $N = \langle c \rangle$ for some $c \in D$. Since $N = \cup_i N_i$, we must have $c \in N_r$, for some $r \in \mathbb{Z}^+$. For $s \geq r$, we have

$$\langle c \rangle \subseteq N_r \subseteq N_s \subseteq N = \langle c \rangle.$$

Thus $N_r = N_s$ for $s \geq r$.

The equivalence with the ACC is immediate. ◆

In what follows, it will be useful to remember that for elements a and b of a domain D,

$\langle a \rangle \subseteq \langle b \rangle$ if and only if b divides a, and

$\langle a \rangle = \langle b \rangle$ if and only if a and b are associates.

For the first property, note that $\langle a \rangle \subseteq \langle b \rangle$ if and only if $a \in \langle b \rangle$, which is true if and only if $a = bd$ for some $d \in D$, so that b divides a. Using this first property, we see that $\langle a \rangle = \langle b \rangle$ if and only if $a = bc$ and $b = ad$ for some $c, d \in D$. But then $a = adc$ and by canceling, we obtain $1 = dc$. Thus d and c are units so a and b are associates.

We can now prove Condition 1 of the definition of a UFD for an integral domain that is a PID.

45.11 Theorem Let D be a PID. Every element that is neither 0 nor a unit in D is a product of irreducibles.

Proof Let $a \in D$, where a is neither 0 nor a unit. We first show that a has at least one irreducible factor. If a is an irreducible, we are done. If a is not an irreducible, then $a = a_1 b_1$, where *neither* a_1 nor b_1 is a unit. Now

$$\langle a \rangle \subset \langle a_1 \rangle,$$

for $\langle a \rangle \subseteq \langle a_1 \rangle$ follows from $a = a_1 b_1$, and if $\langle a \rangle = \langle a_1 \rangle$, then a and a_1 would be associates and b_1 would be a unit, contrary to construction. Continuing this procedure then,

starting now with a_1, we arrive at a strictly ascending chain of ideals

$$\langle a \rangle \subset \langle a_1 \rangle \subset \langle a_2 \rangle \subset \cdots.$$

By the ACC in Lemma 45.10, this chain terminates with some $\langle a_r \rangle$, and a_r must then be irreducible. Thus a has an irreducible factor a_r.

By what we have just proved, for an element a that is neither 0 nor a unit in D, either a is irreducible or $a = p_1 c_1$ for p_1 an irreducible and c_1 not a unit. By an argument similar to the one just made, in the latter case we can conclude that $\langle a \rangle \subset \langle c_1 \rangle$. If c_1 is not irreducible, then $c_1 = p_2 c_2$ for an irreducible p_2 with c_2 not a unit. Continuing, we get a strictly ascending chain of ideals

$$\langle a \rangle \subset \langle c_1 \rangle \subset \langle c_2 \rangle \subset \cdots.$$

This chain must terminate, by the ACC in Lemma 45.10, with some $c_r = q_r$ that is an irreducible. Then $a = p_1 p_2 \cdots p_r q_r$. ◆

This completes our demonstration of Condition 1 of the definition of a UFD. Let us turn to Condition 2. Our arguments here are parallel to those leading to Theorem 23.20. The results we encounter along the way are of some interest in themselves.

45.12 Lemma **(Generalization of Theorem 27.25)** An ideal $\langle p \rangle$ in a PID is maximal if and only if p is an irreducible.

Proof Let $\langle p \rangle$ be a maximal ideal of D, a PID. Suppose that $p = ab$ in D. Then $\langle p \rangle \subseteq \langle a \rangle$. Suppose that $\langle a \rangle = \langle p \rangle$. Then a and p would be associates, so b must be a unit. If $\langle a \rangle \neq \langle p \rangle$, then we must have $\langle a \rangle = \langle 1 \rangle = D$, since $\langle p \rangle$ is maximal. But then a and 1 are associates, so a is a unit. Thus, if $p = ab$, either a or b must be a unit. Hence p is an irreducible of D.

Conversely, suppose that p is an irreducible in D. Then if $\langle p \rangle \subseteq \langle a \rangle$, we must have $p = ab$. Now if a is a unit, then $\langle a \rangle = \langle 1 \rangle = D$. If a is not a unit, then b must be a unit, so there exists $u \in D$ such that $bu = 1$. Then $pu = abu = a$, so $\langle a \rangle \subseteq \langle p \rangle$, and we have $\langle a \rangle = \langle p \rangle$. Thus $\langle p \rangle \subseteq \langle a \rangle$ implies that either $\langle a \rangle = D$ or $\langle a \rangle = \langle p \rangle$, and $\langle p \rangle \neq D$ or p would be a unit. Hence $\langle p \rangle$ is a maximal ideal. ◆

45.13 Lemma **(Generalization of Theorem 27.27)** In a PID, if an irreducible p divides ab, then either $p \mid a$ or $p \mid b$.

Proof Let D be a PID and suppose that for an irreducible p in D we have $p \mid ab$. Then $(ab) \in \langle p \rangle$. Since every maximal ideal in D is a prime ideal by Corollary 27.16, $(ab) \in \langle p \rangle$ implies that either $a \in \langle p \rangle$ or $b \in \langle p \rangle$, giving either $p \mid a$ or $p \mid b$. ◆

45.14 Corollary If p is an irreducible in a PID and p divides the product $a_1 a_2 \cdots a_n$ for $a_i \in D$, then $p \mid a_i$ for at least one i.

Proof Proof of this corollary is immediate from Lemma 45.13 if we use mathematical induction. ◆

45.15 Definition A nonzero nonunit element p of an integral domain D is a **prime** if, for all $a, b \in D$, $p \mid ab$ implies either $p \mid a$ or $p \mid b$. ■

Lemma 45.13 focused our attention on the defining property of a prime. In Exercises 25 and 26, we ask you to show that a prime in an integral domain is always an irreducible and that in a UFD an irreducible is also a prime. Thus the concepts of prime and irreducible coincide in a UFD. Example 45.16 will exhibit an integral domain containing some irreducibles that are not primes, so the concepts do not coincide in every domain.

45.16 Example Let F be a field and let D be the subdomain $F[x^3, xy, y^3]$ of $F[x, y]$. Then x^3, xy, and y^3 are irreducibles in D, but

$$(x^3)(y^3) = (xy)(xy)(xy).$$

Since xy divides x^3y^3 but not x^3 or y^3, we see that xy is not a prime. Similar arguments show that neither x^3 nor y^3 is a prime. ▲

The defining property of a prime is precisely what is needed to establish uniqueness of factorization, Condition 2 in the definition of a UFD. We now complete the proof of Theorem 45.17 by demonstrating the uniqueness of factorization in a PID.

45.17 Theorem **(Generalization of Theorem 23.20)** Every PID is a UFD.

Proof Theorem 45.11 shows that if D is a PID, then each $a \in D$, where a is neither 0 nor a unit, has a factorization

$$a = p_1 p_2 \cdots p_r$$

into irreducibles. It remains for us to show uniqueness. Let

$$a = q_1 q_2 \cdots q_s$$

be another such factorization into irreducibles. Then we have $p_1 \mid (q_1 q_2 \cdots q_s)$, which implies that $p_1 \mid q_j$ for some j by Corollary 45.14. By changing the order of the q_j if necessary, we can assume that $j = 1$ so $p_1 \mid q_1$. Then $q_1 = p_1 u_1$, and since p_1 is an irreducible, u_1 is a unit, so p_1 and q_1 are associates. We have then

$$p_1 p_2 \cdots p_r = p_1 u_1 q_2 \cdots q_s,$$

so by the cancellation law in D,

$$p_2 \cdots p_r = u_1 q_2 \cdots q_s.$$

Continuing this process, starting with p_2 and so on, we finally arrive at

$$1 = u_1 u_2 \cdots u_r q_{r+1} \cdots q_s.$$

Since the q_j are irreducibles, we must have $r = s$. ◆

Example 45.31 at the end of this section will show that the converse to Theorem 45.17 is false. That is, a UFD need not be a PID.

Many algebra texts start by proving the following corollary of Theorem 45.17. We have assumed that you were familiar with this corollary and used it freely in our other work.

45.18 Corollary **(Fundamental Theorem of Arithmetic)** The integral domain $\mathbb{Z}$ is a UFD.

Proof We have seen that all ideals in $\mathbb{Z}$ are of the form $n\mathbb{Z} = \langle n \rangle$ for $n \in \mathbb{Z}$. Thus $\mathbb{Z}$ is a PID. and Theorem 45.17 applies. ◆

It is worth noting that the proof that $\mathbb{Z}$ is a PID was really way back in Corollary 6.7. We proved Theorem 6.6 by using the division algorithm for $\mathbb{Z}$ exactly as we proved, in Theorem 27.24, that $F[x]$ is a PID by using the division algorithm for $F[x]$. In Section 46, we shall examine this parallel more closely.

If D Is a UFD, then $D[x]$ Is a UFD

We now start the proof of Theorem 45.29, our second main result for this section. The idea of the argument is as follows. Let D be a UFD. We can form a field of quotients F of D. Then $F[x]$ is a UFD by Theorem 23.20, and we shall show that we can recover a factorization for $f(x) \in D[x]$ from its factorization in $F[x]$. It will be necessary to compare the irreducibles in $F[x]$ with those in $D[x]$, of course. This approach, which we prefer as more intuitive than some more efficient modern ones, is essentially due to Gauss.

45.19 Definition Let D be a UFD and let $a_1, a_2, \cdots, a_n$ be nonzero elements of D. An element d of D is a **greatest common divisor** (abbreviated gcd) **of all of the** a_i if $d \mid a_i$ for $i = 1, \cdots, n$ and any other $d' \in D$ that divides all the a_i also divides d. ■

In this definition, we called d "a" gcd rather than "the" gcd because gcd's are only defined up to units. Suppose that d and d' are two gcd's of a_i for $i = 1, \cdots, n$. Then $d \mid d'$ and $d' \mid d$ by our definition. Thus $d = q'd'$ and $d' = qd$ for some $q, q' \in D$, so $1d = q'qd$. By cancellation in D, we see that $q'q = 1$ so q and q' are indeed units.

The technique in the example that follows shows that gcd's exist in a UFD.

45.20 Example Let us find a gcd of 420, -168, and 252 in the UFD $\mathbb{Z}$. Factoring, we obtain $420 = 2^2 \cdot 3 \cdot 5 \cdot 7$, $-168 = 2^3 \cdot (-3) \cdot 7$, and $252 = 2^2 \cdot 3^2 \cdot 7$. We choose one of these numbers, say 420, and find the highest power of each of its irreducible factors (up to associates) that divides all the numbers, 420, -168 and 252 in our case. We take as gcd the product of these highest powers of irreducibles. For our example, these powers of irreducible factors of 420 are 2^2, 3^1, 5^0, and 7^1 so we take as gcd $d = 4 \cdot 3 \cdot 1 \cdot 7 = 84$. The only other gcd of these numbers in $\mathbb{Z}$ is -84, because 1 and -1 are the only units. ▲

Execution of the technique in Example 45.20 depends on being able to factor an element of a UFD into a product of irreducibles. This can be a tough job, even in $\mathbb{Z}$. Section 46 will exhibit a technique, the Euclidean Algorithm, that will allow us to find gcd's without factoring in a class of UFD's that includes $\mathbb{Z}$ and $F[x]$ for a field F.

45.21 Definition Let D be a UFD. A nonconstant polynomial

$$f(x) = a_0 + a_1x + \cdots + a_nx^n$$

in $D[x]$ is **primitive** if 1 is a gcd of the a_i for $i = 0, 1, \cdots, n$. ■

45.22 Example In $\mathbb{Z}[x]$, $4x^2 + 3x + 2$ is primitive, but $4x^2 + 6x + 2$ is not, since 2, a nonunit in $\mathbb{Z}$, is a common divisor of 4, 6, and 2. ▲

Observe that every nonconstant irreducible in $D[x]$ must be a primitive polynomial.

45.23 Lemma If D is a UFD, then for every nonconstant $f(x) \in D[x]$ we have $f(x) = (c)g(x)$, where $c \in D$, $g(x) \in D[x]$, and $g(x)$ is primitive. The element c is unique up to a unit factor in D and is the **content of** $f(x)$. Also $g(x)$ is unique up to a unit factor in D.

Proof Let $f(x) \in D[x]$ be given where $f(x)$ is a nonconstant polynomial with coefficients $a_0, a_1, \cdots, a_n$. Let c be a gcd of the a_i for $i = 0, 1, \cdots, n$. Then for each i, we have $a_i = cq_i$ for some $q_i \in D$. By the distributive law, we have $f(x) = (c)g(x)$, where no irreducible in D divides all of the coefficients $q_0, q_1, \cdots, q_n$ of $g(x)$. Thus $g(x)$ is a primitive polynomial.

For uniqueness, if also $f(x) = (d)h(x)$ for $d \in D$, $h(x) \in D[x]$, and $h(x)$ primitive, then each irreducible factor of c must divide d and conversely. By setting $(c)g(x) = (d)h(x)$ and canceling irreducible factors of c into d, we arrive at $(u)g(x) = (v)h(x)$ for a unit $u \in D$. But then v must be a unit of D or we would be able to cancel irreducible factors of v into u. Thus u and v are both units, so c is unique up to a unit factor. From $f(x) = (c)g(x)$, we see that the primitive polynomial $g(x)$ is also unique up to a unit factor. ◆

45.24 Example In $\mathbb{Z}[x]$,

$$4x^2 + 6x - 8 = (2)(2x^2 + 3x - 4),$$

where $2x^3 + 3x - 4$ is primitive. ▲

45.25 Lemma **(Gauss's Lemma)** If D is a UFD, then a product of two primitive polynomials in $D[x]$ is again primitive.

Proof Let

$$f(x) = a_0 + a_1x + \cdots + a_nx^n$$

and

$$g(x) = b_0 + b_1x + \cdots + b_mx^m$$

be primitive in $D[x]$, and let $h(x) = f(x)g(x)$. Let p be an irreducible in D. Then p does not divide all a_i and p does not divide all b_j, since $f(x)$ and $g(x)$ are primitive. Let a_r be the first coefficient of $f(x)$ not divisible by p; that is, $p \mid a_i$ for $i < r$, but $p \nmid a_r$ (that is, p does not divide a_r). Similarly, let $p \mid b_j$ for $j < s$, but $p \nmid b_s$. The coefficient of x^{r+s} in $h(x) = f(x)g(x)$ is

$$c_{r+s} = (a_0b_{r+s} + \cdots + a_{r-1}b_{s+1}) + a_rb_s + (a_{r+1}b_{s-1} + \cdots + a_{r+s}b_0).$$

Now $p \mid a_i$ for $i < r$ implies that

$$p \mid (a_0 b_{r+s} + \cdots + a_{r-1} b_{s+1}),$$

and also $p \mid b_j$ for $j < s$ implies that

$$p \mid (a_{r+1} b_{s-1} + \cdots + a_{r+s} b_0).$$

But p does not divide a_r or b_s, so p does not divide $a_r b_s$, and consequently p does not divide c_{r+s}. This shows that given an irreducible $p \in D$, there is some coefficient of $f(x)g(x)$ not divisible by p. Thus $f(x)g(x)$ is primitive. ◆

45.26 Corollary If D is a UFD, then a finite product of primitive polynomials in $D[x]$ is again primitive.

Proof This corollary follows from Lemma 45.25 by induction. ◆

Now let D be a UFD and let F be a field of quotients of D. By Theorem 23.20, $F[x]$ is a UFD. As we said earlier, we shall show that $D[x]$ is a UFD by carrying a factorization in $F[x]$ of $f(x) \in D[x]$ back into one in $D[x]$. The next lemma relates the nonconstant irreducibles of $D[x]$ to those of $F[x]$. This is the last important step.

45.27 Lemma Let D be a UFD and let F be a field of quotients of D. Let $f(x) \in D[x]$, where (degree $f(x)) > 0$. If $f(x)$ is an irreducible in $D[x]$, then $f(x)$ is also an irreducible in $F[x]$. Also, if $f(x)$ is primitive in $D[x]$ and irreducible in $F[x]$, then $f(x)$ is irreducible in $D[x]$.

Proof Suppose that a nonconstant $f(x) \in D[x]$ factors into polynomials of lower degree in $F[x]$, that is,

$$f(x) = r(x)s(x)$$

for $r(x), s(x) \in F[x]$. Then since F is a field of quotients of D, each coefficient in $r(x)$ and $s(x)$ is of the form a/b for some $a, b \in D$. By clearing denominators, we can get

$$(d)f(x) = r_1(x)s_1(x)$$

for $d \in D$, and $r_1(x), s_1(x) \in D[x]$, where the degrees of $r_1(x)$ and $s_1(x)$ are the degrees of $r(x)$ and $s(x)$, respectively. By Lemma 45.23, $f(x) = (c)g(x)$, $r_1(x) = (c_1)r_2(x)$, and $s_1(x) = (c_2)s_2(x)$ for primitive polynomials $g(x)$, $r_2(x)$, and $s_2(x)$, and $c, c_1, c_2 \in D$. Then

$$(dc)g(x) = (c_1c_2)r_2(x)s_2(x),$$

and by Lemma 45.25, $r_2(x)s_2(x)$ is primitive. By the uniqueness part of Lemma 45.23, $c_1c_2 = dcu$ for some unit u in D. But then

$$(dc)g(x) = (dcu)r_2(x)s_2(x),$$

so

$$f(x) = (c)g(x) = (cu)r_2(x)s_2(x).$$

We have shown that if $f(x)$ factors nontrivially in $F[x]$, then $f(x)$ factors nontrivially into polynomials of the same degrees in $D[x]$. Thus if $f(x) \in D[x]$ is irreducible in $D[x]$, it must be irreducible in $F[x]$.

A nonconstant $f(x) \in D[x]$ that is primitive in $D[x]$ and irreducible in $F[x]$ is also irreducible in $D[x]$, since $D[x] \subseteq F[x]$. ◆

Lemma 45.27 shows that if D is a UFD, the irreducibles in $D[x]$ are precisely the irreducibles in D, together with the nonconstant primitive polynomials that are irreducible in $F[x]$, where F is a field of quotients of $D[x]$.

The preceding lemma is very important in its own right. This is indicated by the following corollary, a special case of which was our Theorem 23.11. (We admit that it does not seem very sensible to call a special case of a corollary of a lemma a theorem. The label assigned to a result depends somewhat on the context in which it appears.)

45.28 Corollary If D is a UFD and F is a field of quotients of D, then a nonconstant $f(x) \in D[x]$ factors into a product of two polynomials of lower degrees r and s in $F[x]$ if and only if it has a factorization into polynomials of the same degrees r and s in $D[x]$.

Proof It was shown in the proof of Lemma 45.27 that if $f(x)$ factors into a product of two polynomials of lower degree in $F[x]$, then it has a factorization into polynomials of the same degrees in $D[x]$ (see the next to last sentence of the first paragraph of the proof).

The converse holds since $D[x] \subseteq F[x]$. ◆

We are now prepared to prove our main theorem.

45.29 Theorem If D is a UFD, then $D[x]$ is a UFD.

Proof Let $f(x) \in D[x]$, where $f(x)$ is neither 0 nor a unit. If $f(x)$ is of degree 0, we are done, since D is a UFD. Suppose that (degree $f(x)$) > 0. Let

$$f(x) = g_1(x)g_2(x)\cdots g_r(x)$$

be a factorization of $f(x)$ in $D[x]$ having the greatest number r of factors of positive degree. (There is such a greatest number of such factors because r cannot exceed the degree of $f(x)$.) Now factor each $g_i(x)$ in the form $g_i(x) = c_i h_i(x)$ where c_i is the content of $g_i(x)$ and $h_i(x)$ is a primitive polynomial. Each of the $h_i(x)$ is irreducible, because if it could be factored, none of the factors could lie in D, hence all would have positive degree leading to a corresponding factorzation of $g_i(x)$, and then to a factorization of $f(x)$ with more than r factors of positive degree, contradicting our choice of r. Thus we now have

$$f(x) = c_1h_1(x)c_2h_2(x)\cdots c_rh_r(x)$$

where the $h_i(x)$ are irreducible in $D[x]$. If we now factor the c_i into irreducibles in D, we obtain a factorization of $f(x)$ into a product of irreducibles in $D[x]$.

The factorization of $f(x) \in D[x]$, where $f(x)$ has degree 0, is unique since D is a UFD; see the comment following Lemma 45.27. If $f(x)$ has degree greater than 0, we

can view any factorization of $f(x)$ into irreducibles in $D[x]$ as a factorization in $F[x]$ into units (that is, the factors in D) and irreducible polynomials in $F[x]$ by Lemma 45.27. By Theorem 23.20, these polynomials are unique, except for possible constant factors in F. But as an irreducible in $D[x]$, each polynomial of degree >0 appearing in the factorization of $f(x)$ in $D[x]$ is primitive. By the uniqueness part of Lemma 45.23, this shows that these polynomials are unique in $D[x]$ up to unit factors, that is, associates. The product of the irreducibles in D in the factorization of $f(x)$ is the content of $f(x)$, which is again unique up to a unit factor by Lemma 45.23. Thus all irreducibles in $D[x]$ appearing in the factorization are unique up to order and associates. ◆

45.30 Corollary If F is a field and $x_1, \cdots, x_n$ are indeterminates, then $F[x_1, \cdots, x_n]$ is a UFD.

Proof By Theorem 23.20, $F[x_1]$ is a UFD. By Theorem 45.29, so is $(F[x_1])[x_2] = F[x_1, x_2]$. Continuing in this procedure, we see (by induction) that $F[x_1, \cdots, x_n]$ is a UFD. ◆

We have seen that a PID is a UFD. Corollary 45.30 makes it easy for us to give an example that shows that *not every* UFD *is a* PID.

45.31 Example Let F be a field and let x and y be indeterminates. Then $F[x, y]$ is a UFD by Corollary 45.30. Consider the set N of all polynomials in x and y in $F[x, y]$ having constant term 0. Then N is an ideal, but not a principal ideal. Thus $F[x, y]$ is not a PID. ▲

Another example of a UFD that is not a PID is $\mathbb{Z}[x]$, as shown in Exercise 12, Section 46.

EXERCISES 45

Computations

In Exercises 1 through 8, determine whether the element is an irreducible of the indicated domain.

1. 5 in $\mathbb{Z}$

2. -17 in $\mathbb{Z}$

3. 14 in $\mathbb{Z}$

4. $2x - 3$ in $\mathbb{Z}[x]$

5. $2x - 10$ in $\mathbb{Z}[x]$

6. $2x - 3$ in $\mathbb{Q}[x]$

7. $2x - 10$ in $\mathbb{Q}[x]$

8. $2x - 10$ in $\mathbb{Z}_{11}[x]$

9. If possible, give four different associates of $2x - 7$ viewed as an element of $\mathbb{Z}[x]$; of $\mathbb{Q}[x]$; of $\mathbb{Z}_{11}[x]$.

10. Factor the polynomial $4x^2 - 4x + 8$ into a product of irreducibles viewing it as an element of the integral domain $\mathbb{Z}[x]$; of the integral domain $\mathbb{Q}[x]$; of the integral domain $\mathbb{Z}_{11}[x]$.

In Exercises 11 through 13, find all gcd's of the given elements of $\mathbb{Z}$.

11. 234, 3250, 1690

12. 784, -1960, 448

13. 2178, 396, 792, 594

In Exercises 14 through 17, express the given polynomial as the product of its content with a primitive polynomial in the indicated UFD.

14. $18x^2 - 12x + 48$ in $\mathbb{Z}[x]$

15. $18x^2 - 12x + 48$ in $\mathbb{Q}[x]$

16. $2x^2 - 3x + 6$ in $\mathbb{Z}[x]$

17. $2x^2 - 3x + 6$ in $\mathbb{Z}_7[x]$

Concepts

In Exercises 18 through 20, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

18. Two elements a and b in an integral domain D are *associates* in D if and only if their quotient a/b in D is a unit.

19. An element of an integral domain D is an *irreducible* of D if and only if it cannot be factored into a product of two elements of D.

20. An element of an integral domain D is a *prime* of D if and only if it cannot be factored into a product of two smaller elements of D.

21. Mark each of the following true or false.

_____ **a.** Every field is a UFD.
_____ **b.** Every field is a PID.
_____ **c.** Every PID is a UFD.
_____ **d.** Every UFD is a PID.
_____ **e.** $\mathbb{Z}[x]$ is a UFD.
_____ **f.** Any two irreducibles in any UFD are associates.
_____ **g.** If D is a PID, then $D[x]$ is a PID.
_____ **h.** If D is a UFD, then $D[x]$ is a UFD.
_____ **i.** In any UFD, if $p \mid a$ for an irreducible p, then p itself appears in every factorization of a.
_____ **j.** A UFD has no divisors of 0.

22. Let D be a UFD. Describe the irreducibles in $D[x]$ in terms of the irreducibles in D and the irreducibles in $F[x]$, where F is a field of quotients of D.

23. Lemma 45.26 states that if D is a UFD with a field of quotients F, then a nonconstant irreducible $f(x)$ of $D[x]$ is also an irreducible of $F[x]$. Show by an example that a $g(x) \in D[x]$ that is an irreducible of $F[x]$ need not be an irreducible of $D[x]$.

24. All our work in this section was restricted to integral domains. Taking the same definition in this section but for a commutative ring with unity, consider factorizations into irreducibles in $\mathbb{Z} \times \mathbb{Z}$. What can happen? Consider in particular $(1, 0)$.

Theory

25. Prove that if p is a prime in an integral domain D, then p is an irreducible.

26. Prove that if p is an irreducible in a UFD, then p is a prime.

27. For a commutative ring R with unity show that the relation $a \sim b$ if a is an associate of b (that is, if $a = bu$ for u a unit in R is an equivalence relation on R.

28. Let D be an integral domain. Exercise 37, Section 18 showed that $\langle U, \cdot \rangle$ is a group where U is the set of units of D. Show that the set $D^* - U$ of nonunits of D excluding 0 is closed under multiplication. Is this set a group under the multiplication of D?

29. Let D be a UFD. Show that a nonconstant divisor of a primitive polynomial in $D[x]$ is again a primitive polynomial.

30. Show that in a PID, every ideal is contained in a maximal ideal. [*Hint:* Use Lemma 45.10.]

31. Factor $x^3 - y^3$ into irreducibles in $\mathbb{Q}[x, y]$ and prove that each of the factors is irreducible.

There are several other concepts often considered that are similar in character to the ascending chain condition on ideals in a ring. The following three exercises concern some of these concepts.

32. Let R be any ring. The **ascending chain condition** (ACC) **for ideals** holds in R if every strictly increasing sequence $N_1 \subset N_2 \subset N_3 \subset \cdots$ of ideals in R is of finite length. The **maximum condition** (MC) **for ideals** holds in R if every nonempty set S of ideals in R contains an ideal not properly contained in any other ideal of the set S. The **finite basis condition** (FBC) **for ideals** holds in R if for each ideal N in R, there is a finite set $B_N = \{b_1, \cdots, b_n\} \subseteq N$ such that N is the intersection of all ideals of R containing B_N. The set B_N is a **finite generating set for** N.

Show that for every ring R, the conditions ACC, MC, and FBC are equivalent.

33. Let R be any ring. The **descending chain condition** (DCC) **for ideals** holds in R if every strictly decreasing sequence $N_1 \supset N_2 \supset N_3 \supset \cdots$ of ideals in R is of finite length. The **minimum condition** (mC) **for ideals** holds in R if given any set S of ideals of R, there is an ideal of S that does not properly contain any other ideal in the set S.

Show that for every ring, the conditions DCC and mC are equivalent.

34. Give an example of a ring in which ACC holds but DCC does not hold. (See Exercises 32 and 33.)

SECTION 46 EUCLIDEAN DOMAINS

We have remarked several times on the importance of division algorithms. Our first contact with them was the *division algorithm for* $\mathbb{Z}$ in Section 6. This algorithm was immediately used to prove the important theorem that a subgroup of a cyclic group is cyclic, that is, has a single generator. Of course, this shows at once that $\mathbb{Z}$ is a PID. The *division algorithm for* $F[x]$ appeared in Theorem 23.1 and was used in a completely analogous way to show that $F[x]$ is a PID. Now a modern technique of mathematics is to take some clearly related situations and to try to bring them under one roof by abstracting the important ideas common to them. The following definition is an illustration of this technique, as is this whole text! Let us see what we can develop by starting with the existence of a fairly general division algorithm in an integral domain.

46.1 Definition A **Euclidean norm** on an integral domain D is a function ν mapping the nonzero elements of D into the nonnegative integers such that the following conditions are satisfied:

1. For all $a, b \in D$ with $b \neq 0$, there exist q and r in D such that $a = bq + r$, where either $r = 0$ or $\nu(r) < \nu(b)$.
2. For all $a, b \in D$, where neither a nor b is 0, $\nu(a) \leq \nu(ab)$.

An integral domain D is a **Euclidean domain** if there exists a Euclidean norm on D. ■

The importance of Condition 1 is clear from our discussion. The importance of Condition 2 is that it will enable us to characterize the units of a Euclidean domain D.

46.2 Example The integral domain $\mathbb{Z}$ is a Euclidean domain, for the function ν defined by $\nu(n) = |n|$ for $n \neq 0$ in $\mathbb{Z}$ is a Euclidean norm on $\mathbb{Z}$. Condition 1 holds by the division algorithm for $\mathbb{Z}$. Condition 2 follows from $|ab| = |a||b|$ and $|a| \geq 1$ for $a \neq 0$ in $\mathbb{Z}$. ▲

46.3 Example If F is a field, then $F[x]$ is a Euclidean domain, for the function ν defined by $\nu(f(x)) =$ (degree $f(x)$) for $f(x) \in F[x]$, and $f(x) \neq 0$ is a Euclidean norm. Condition 1 holds by Theorem 23.1, and Condition 2 holds since the degree of the product of two polynomials is the sum of their degrees. ▲

Of course, we should give some examples of Euclidean domains other than these familiar ones that motivated the definition. We shall do this in Section 47. In view of the opening remarks, we anticipate the following theorem.

46.4 Theorem Every Euclidean domain is a PID.

Proof Let D be a Euclidean domain with a Euclidean norm ν, and let N be an ideal in D. If $N = \{0\}$, then $N = \langle 0 \rangle$ and N is principal. Suppose that $N \neq \{0\}$. Then there exists $b \neq 0$ in N. Let us choose b such that $\nu(b)$ is minimal among all $\nu(n)$ for $n \in N$. We claim that $N = \langle b \rangle$. Let $a \in N$. Then by Condition 1 for a Euclidean domain, there exist q and r in D such that

$$a = bq + r,$$

where either $r = 0$ or $\nu(r) < \nu(b)$. Now $r = a - bq$ and $a, b \in N$, so that $r \in N$ since N is an ideal. Thus $\nu(r) < \nu(b)$ is impossible by our choice of b. Hence $r = 0$, so $a = bq$. Since a was any element of N, we see that $N = \langle b \rangle$. ◆

46.5 Corollary A Euclidean domain is a UFD.

Proof By Theorem 46.4, a Euclidean domain is a PID and by Theorem 45.17, a PID is a UFD. ◆

Finally, we should mention that while a Euclidean domain is a PID by Theorem 46.4, not every PID is a Euclidean domain. Examples of PIDs that are not Euclidean are not easily found, however.

Arithmetic in Euclidean Domains

We shall now investigate some properties of Euclidean domains related to their multiplicative structure. We emphasize that the arithmetic structure of a Euclidean domain is not affected in any way by a Euclidean norm ν on the domain. A Euclidean norm is merely a useful tool for possibly throwing some light on this arithmetic structure of the domain. The arithmetic structure of a domain D is completely determined by the set D and the two binary operations $+$ and $\cdot$ on D.

Let D be a Euclidean domain with a Euclidean norm ν. We can use Condition 2 of a Euclidean norm to characterize the units of D.

46.6 Theorem For a Euclidean domain with a Euclidean norm ν, $\nu(1)$ is minimal among all $\nu(a)$ for nonzero $a \in D$, and $u \in D$ is a unit if and only if $\nu(u) = \nu(1)$.

Proof Condition 2 for ν tells us at once that for $a \neq 0$,

$$\nu(1) \leq \nu(1a) = \nu(a).$$

On the other hand, if u is a unit in D, then

$$\nu(u) \leq \nu(uu^{-1}) = \nu(1).$$

Thus

$$\nu(u) = \nu(1)$$

for a unit u in D.

Conversely, suppose that a nonzero $u \in D$ is such that $v(u) = v(1)$. Then by the division algorithm, there exist q and r in D such that

$$1 = uq + r,$$

where either $r = 0$ or $v(r) < v(u)$. But since $v(u) = v(1)$ is minimal over all $v(d)$ for nonzero $d \in D$, $v(r) < v(u)$ is impossible. Hence $r = 0$ and $1 = uq$, so u is a unit. ◆

46.7 Example For $\mathbb{Z}$ with $v(n) = |n|$, the minimum of $v(n)$ for nonzero $n \in \mathbb{Z}$ is 1, and 1 and -1 are the only elements of $\mathbb{Z}$ with $v(n) = 1$. Of course, 1 and -1 are exactly the units of $\mathbb{Z}$. ▲

46.8 Example For $F[x]$ with $v(f(x)) = (\text{degree } f(x))$ for $f(x) \neq 0$, the minimum value of $v(f(x))$ for all nonzero $f(x) \in F[x]$ is 0. The nonzero polynomials of degree 0 are exactly the nonzero elements of F, and these are precisely the units of $F[x]$. ▲

We emphasize that everything we prove here holds in *every* Euclidean domain, in particular in $\mathbb{Z}$ and $F[x]$. As indicated in Example 45.20, we can show that any a and b in a UFD have a gcd and actually compute one by factoring a and b into irreducibles, but such factorizations can be very tough to find. However, if a UFD is actually Euclidean, and we know an easily computed Euclidean norm, there is an easy constructive way to find gcd's, as the next theorem shows.

Historical Note

The Euclidean algorithm appears in Euclid's *Elements* as propositions 1 and 2 of Book VII, where it is used as here to find the greatest common divisor of two integers. Euclid uses it again in Book X (propositions 2 and 3) to find the greatest common measure of two magnitudes (if it exists) and to determine whether two magnitudes are incommensurable.

The algorithm appears again in the *Brahmesphutasiddhanta* (Correct Astronomical System of Brahma) (628) of the seventh-century Indian mathematician and astronomer Brahmagupta. To solve the indeterminate equation $rx + c = sy$ in integers, Brahmagupta uses Euclid's procedure to "reciprocally divide" r by s until he reaches the final nonzero remainder. By then using, in effect, a substitution procedure based on the various quotients and remainders, he produces a straightforward algorithm for finding the smallest positive solution to his equation.

The thirteenth-century Chinese algebraist Qin Jiushao also used the Euclidean algorithm in his solution of the so-called Chinese Remainder problem published in the *Shushu jiuzhang* (Mathematical Treatise in Nine Sections) (1247). Qin's goal was to display a method for solving the system of congruences $N \equiv r_i \pmod{m_i}$. As part of that method he needed to solve congruences of the form $Nx \equiv 1 \pmod{m}$, where N and m are relatively prime. The solution to a congruence of this form is again found by a substitution procedure, different from the Indian one, using the quotients and remainders from the Euclidean algorithm applied to N and m. It is not known whether the common element in the Indian and Chinese algorithms, the Euclidean algorithm itself, was discovered independently in these cultures or was learned from Greek sources.

46.9 Theorem **(Euclidean Algorithm)** Let D be a Euclidean domain with a Euclidean norm ν, and let a and b be nonzero elements of D. Let r_1 be as in Condition 1 for a Euclidean norm, that is,

$$a = bq_1 + r_1,$$

where either $r_1 = 0$ or $\nu(r_1) < \nu(b)$. If $r_1 \neq 0$, let r_2 be such that

$$b = r_1 q_2 + r_2,$$

where either $r_2 = 0$ or $\nu(r_2) < \nu(r_1)$. In general, let r_{i+1} be such that

$$r_{i-1} = r_i q_{i+1} + r_{i+1},$$

where either $r_{i+1} = 0$ or $\nu(r_{i+1}) < \nu(r_i)$. Then the sequence $r_i, r_2, \cdots$ must terminate with some $r_s = 0$. If $r_1 = 0$, then b is a gcd of a and b. If $r_1 \neq 0$ and r_s is the first $r_i = 0$, then a gcd of a and b is r_{s-1}.

Furthermore, if d is a gcd of a and b, then there exist λ and μ in D such that $d = \lambda a + \mu b$.

Proof Since $\nu(r_i) < \nu(r_{i-1})$ and $\nu(r_i)$ is a nonnegative integer, it follows that after some finite number of steps we must arrive at some $r_s = 0$.

If $r_1 = 0$, then $a = bq_1$, and b is a gcd of a and b. Suppose $r_1 \neq 0$. Then if $d \mid a$ and $d \mid b$, we have

$$d \mid (a - bq_1),$$

so $d \mid r_1$. However, if $d_1 \mid r_1$ and $d_1 \mid b$, then

$$d_1 \mid (bq_1 + r_1),$$

so $d_1 \mid a$. Thus the set of common divisors of a and b is the same set as the set of common divisors of b and r_1. By a similar argument, if $r_2 \neq 0$, the set of common divisors of b and r_1 is the same set as the set of common divisors of r_1 and r_2. Continuing this process, we see finally that the set of common divisors of a and b is the same set as the set of common divisors of r_{s-2} and r_{s-1}, where r_s is the first r_i equal to 0. Thus a gcd of r_{s-2} and r_{s-1} is also a gcd of a and b. But the equation

$$r_{s-2} = q_s r_{s-1} + r_s = q_s r_{s-1}$$

shows that a gcd of r_{s-2} and r_{s-1} is r_{s-1}.

It remains to show that we can express a gcd d of a and b as $d = \lambda a + \mu b$. In terms of the construction just given, if $d = b$, then $d = 0a + 1b$ and we are done. If $d = r_{s-1}$, then, working backward through our equations, we can express each r_i in the form $\lambda_i r_{i-1} + \mu_i r_{i-2}$ for some $\lambda_i, \mu_i \in D$. To illustrate using the first step, from the equation

$$r_{s-3} = q_{s-1} r_{s-2} + r_{s-1}$$

we obtain

$$d = r_{s-1} = r_{s-3} - q_{s-1} r_{s-2}. \tag{1}$$

We then express r_{s-2} in terms of r_{s-3} and r_{s-4} and substitute in Eq. (1) to express d in terms of r_{s-3} and r_{s-4}. Eventually, we will have

$$\begin{aligned} d &= \lambda_3 r_2 + \mu_3 r_1 = \lambda_3(b - r_1 q_2) + \mu_3 r_1 = \lambda_3 b + (\mu_3 - \lambda_3 q_2) r_1 \\ &= \lambda_3 b + (\mu_3 - \lambda_3 q_2)(a - b q_1) \end{aligned}$$

which can be expressed in the form $d = \lambda a + \mu b$. If d' is any other gcd of a and b, then $d' = ud$ for some unit u, so $d' = (\lambda u)a + (\mu u)b$. ◆

The nice thing about Theorem 46.9 is that it can be implemented on a computer. Of course, we anticipate that of anything that is labeled an "algorithm."

46.10 Example Let us illustrate the Euclidean algorithm for the Euclidean norm $|\ |$ on $\mathbb{Z}$ by computing a gcd of 22,471 and 3,266. We just apply the division algorithm over and over again, and the last nonzero remainder is a gcd. We label the numbers obtained as in Theorem 46.9 to further illustrate the statement and proof of the theorem. The computations are easily checked.

$$\begin{aligned} & & a &= 22{,}471 \\ & & b &= 3{,}266 \\ 22{,}471 &= (3{,}266)6 + 2{,}875 & r_1 &= 2{,}875 \\ 3{,}266 &= (2{,}875)1 + 391 & r_2 &= 391 \\ 2{,}875 &= (391)7 + 138 & r_3 &= 138 \\ 391 &= (138)2 + 115 & r_4 &= 115 \\ 138 &= (115)1 + 23 & r_5 &= 23 \\ 115 &= (23)5 + 0 & r_6 &= 0 \end{aligned}$$

Thus $r_5 = 23$ is a gcd of 22,471 and 3,266. We found a gcd without factoring! This is important, for sometimes it is very difficult to find a factorization of an integer into primes. ▲

46.11 Example Note that the division algorithm Condition 1 in the definition of a Euclidean norm says nothing about r being "positive." In computing a gcd in $\mathbb{Z}$ by the Euclidean algorithm for $|\ |$, as in Example 46.10, it is surely to our interest to make $|r_i|$ as small as possible in each division. Thus, repeating Example 46.10, it would be more efficient to write

$$\begin{aligned} & & a &= 22{,}471 \\ & & b &= 3{,}266 \\ 22{,}471 &= (3{,}266)7 - 391 & r_1 &= -391 \\ 3{,}266 &= (391)8 + 138 & r_2 &= 138 \\ 391 &= (138)3 - 23 & r_3 &= -23 \\ 138 &= (23)6 + 0 & r_4 &= 0 \end{aligned}$$

We can change the sign of r_i from negative to positive when we wish since the divisors of r_i and $-r_i$ are the same. ▲

EXERCISES 46

Computations

In Exercises 1 through 5, state whether the given function ν is a Euclidean norm for the given integral domain.

1. The function ν for $\mathbb{Z}$ given by $\nu(n) = n^2$ for nonzero $n \in \mathbb{Z}$
2. The function ν for $\mathbb{Z}[x]$ given by $\nu(f(x)) =$ (degree of $f(x)$) for $f(x) \in \mathbb{Z}[x]$, $f(x) \neq 0$
3. The function ν for $\mathbb{Z}[x]$ given by $\nu(f(x)) =$ (the absolute value of the coefficient of the highest degree nonzero term of $f(x)$) for nonzero $f(x) \in \mathbb{Z}[x]$
4. The function ν for $\mathbb{Q}$ given by $\nu(a) = a^2$ for nonzero $a \in \mathbb{Q}$
5. The function ν for $\mathbb{Q}$ given by $\nu(a) = 50$ for nonzero $a \in \mathbb{Q}$
6. By referring to Example 46.11, actually express the gcd 23 in the form $\lambda(22{,}471) + \mu(3{,}266)$ for $\lambda, \mu \in \mathbb{Z}$. [*Hint:* From the next to the last line of the computation in Example 46.11, $23 = (138)3 - 391$. From the line before that, $138 = 3{,}266 - (391)8$, so substituting, you get $23 = [3{,}266 - (391)8]3 - 391$, and so on. That is, work your way back up to actually find values for λ and μ.]
7. Find a gcd of 49,349 and 15,555 in $\mathbb{Z}$.
8. Following the idea of Exercise 6 and referring to Exercise 7, express the positive gcd of 49,349 and 15,555 in $\mathbb{Z}$ in the form $\lambda(49{,}349) + \mu(15{,}555)$ for $\lambda, \mu \in \mathbb{Z}$.
9. Find a gcd of

$$x^{10} - 3x^9 + 3x^8 - 11x^7 + 11x^6 - 11x^5 + 19x^4 - 13x^3 + 8x^2 - 9x + 3$$

and

$$x^6 - 3x^5 + 3x^4 - 9x^3 + 5x^2 - 5x + 2$$

in $\mathbb{Q}[x]$.

10. Describe how the Euclidean Algorithm can be used to find the gcd of n members $a_1, a_2, \cdots, a_n$ of a Euclidean domain.
11. Using your method devised in Exercise 10, find the gcd of 2178, 396, 792, and 726.

Concepts

12. Let us consider $\mathbb{Z}[x]$.
    - **a.** Is $\mathbb{Z}[x]$ a UFD? Why?
    - **b.** Show that $\{a + xf(x) \mid a \in 2\mathbb{Z}, f(x) \in \mathbb{Z}[x]\}$ is an ideal in $\mathbb{Z}[x]$.
    - **c.** Is $\mathbb{Z}[x]$ a PID? (Consider part (b).)
    - **d.** Is $\mathbb{Z}[x]$ a Euclidean domain? Why?
13. Mark each of the following true or false.
    - _____ **a.** Every Euclidean domain is a PID.
    - _____ **b.** Every PID is a Euclidean domain.
    - _____ **c.** Every Euclidean domain is a UFD.
    - _____ **d.** Every UFD is a Euclidean domain.
    - _____ **e.** A gcd of 2 and 3 in $\mathbb{Q}$ is $\frac{1}{2}$.
    - _____ **f.** The Euclidean algorithm gives a constructive method for finding a gcd of two integers.
    - _____ **g.** If ν is a Euclidean norm on a Euclidean domain D, then $\nu(1) \leq \nu(a)$ for all nonzero $a \in D$.

_____ **h.** If ν is a Euclidean norm on a Euclidean domain D, then $\nu(1) < \nu(a)$ for all nonzero $a \in D$, $a \neq 1$.

_____ **i.** If ν is a Euclidean norm on a Euclidean domain D, then $\nu(1) < \nu(a)$ for all nonzero nonunits $a \in D$.

_____ **j.** For any field F, $F[x]$ is a Euclidean domain.

14. Does the choice of a particular Euclidean norm ν on a Euclidean domain D influence the arithmetic structure of D in any way? Explain.

Theory

15. Let D be a Euclidean domain and let ν be a Euclidean norm on D. Show that if a and b are associates in D, then $\nu(a) = \nu(b)$.

16. Let D be a Euclidean domain and let ν be a Euclidean norm on D. Show that for nonzero $a, b \in D$, one has $\nu(a) < \nu(ab)$ if and only if b is not a unit of D. [*Hint:* Argue from Exercise 15 that $\nu(a) < \nu(ab)$ implies that b is not a unit of D. Using the Euclidean algorithm, show that $\nu(a) = \nu(ab)$ implies $\langle a \rangle = \langle ab \rangle$. Conclude that if b is not a unit, then $\nu(a) < \nu(ab)$.]

17. Prove or disprove the following statement: If ν is a Euclidean norm on Euclidean domain D, then $\{a \in D \mid \nu(a) > \nu(1)\} \cup \{0\}$ is an ideal of D.

18. Show that every field is a Euclidean domain.

19. Let ν be a Euclidean norm on a Euclidean domain D.

a. Show that if $s \in \mathbb{Z}$ such that $s + \nu(1) > 0$, then $\eta : D^* \to \mathbb{Z}$ defined by $\eta(a) = \nu(a) + s$ for nonzero $a \in D$ is a Euclidean norm on D. As usual, D^* is the set of nonzero elements of D.

b. Show that for $t \in \mathbb{Z}^+$, $\lambda : D^* \to \mathbb{Z}$ given by $\lambda(a) = t \cdot \nu(a)$ for nonzero $a \in D$ is a Euclidean norm on D.

c. Show that there exists a Euclidean norm μ on D such that $\mu(1) = 1$ and $\mu(a) > 100$ for all nonzero nonunits $a \in D$.

20. Let D be a UFD. An element c in D is a **least common multiple** (abbreviated lcm) of two elements a and b in D if $a \mid c$, $b \mid c$ and if c divides every element of D that is divisible by both a and b. Show that every two nonzero elements a and b of a Euclidean domain D have an lcm in D. [*Hint:* Show that all common multiples, in the obvious sense, of both a and b form an ideal of D.]

21. Use the last statement in Theorem 46.9 to show that two nonzero elements $r, s \in \mathbb{Z}$ generate the group $\langle \mathbb{Z}, + \rangle$ if and only if r and s, viewed as integers in the domain $\mathbb{Z}$, are **relatively prime,** that is, have a gcd of 1.

22. Using the last statement in Theorem 46.9, show that for nonzero $a, b, n \in \mathbb{Z}$, the congruence $ax \equiv b \pmod{n}$ has a solution in $\mathbb{Z}$ if a and n are relatively prime.

23. Generalize Exercise 22 by showing that for nonzero $a, b, n \in \mathbb{Z}$, the congruence $ax \equiv b \pmod{n}$ has a solution in $\mathbb{Z}$ if and only if the positive gcd of a and n in $\mathbb{Z}$ divides b. Interpret this result in the ring $\mathbb{Z}_n$.

24. Following the idea of Exercises 6 and 23, outline a constructive method for finding a solution in $\mathbb{Z}$ of the congruence $ax \equiv b \pmod{n}$ for nonzero $a, b, n \in \mathbb{Z}$, if the congruence does have a solution. Use this method to find a solution of the congruence $22x \equiv 18 \pmod{42}$.

SECTION 47 GAUSSIAN INTEGERS AND MULTIPLICATIVE NORMS

Gaussian Integers

We should give an example of a Euclidean domain different from $\mathbb{Z}$ and $F[x]$.

47.1 Definition **A Gaussian integer** is a complex number $a + bi$, where $a, b \in \mathbb{Z}$. For a Gaussian integer $\alpha = a + bi$, the **norm** $N(\alpha)$ **of** α is $a^2 + b^2$. ■

We shall let $\mathbb{Z}[i]$ be the set of all Gaussian integers. The following lemma gives some basic properties of the norm function N on $\mathbb{Z}[i]$ and leads to a demonstration that the function ν defined by $\nu(\alpha) = N(\alpha)$ for nonzero $\alpha \in \mathbb{Z}[i]$ is a Euclidean norm on $\mathbb{Z}[i]$. Note that the Gaussian integers include all the **rational integers,** that is, all the elements of $\mathbb{Z}$.

Historical Note

In his *Disquisitiones Arithmeticae*, Gauss studied in detail the theory of quadratic residues, that is, the theory of solutions to the congruence $x^2 \equiv p \pmod{q}$ and proved the famous quadratic reciprocity theorem showing the relationship between the solutions of the congruences $x^2 \equiv p \pmod{q}$ and $x^2 \equiv q \pmod{p}$ where p and q are primes. In attempting to generalize his results to theories of quartic residues, however, Gauss realized that it was much more natural to consider the Gaussian integers rather than the ordinary integers.

Gauss's investigations of the Gaussian integers are contained in a long paper published in 1832 in which he proved various analogies between them and the ordinary integers. For example, after noting that there are four units (invertible elements) among the Gaussian integers, namely $1, -1, i$, and $-i$, and defining the norm as in Definition 47.1, he generalized the notion of a prime integer by defining a prime Gaussian integer to be one that cannot be expressed as the product of two other integers, neither of them units. He was then able to determine which Gaussian integers are prime: A Gaussian integer that is not real is prime if and only if its norm is a real prime, which can only be 2 or of the form $4n + 1$. The real prime $2 = (1 + i)(1 - i)$ and real primes congruent to 1 modulo 4 like $13 = (2 + 3i)(2 - 3i)$ factor as the product of two Gaussian primes. Real primes of the form $4n + 3$ like 7 and 11 are still prime in the domain of Gaussian integers. See Exercise 10.

47.2 Lemma In $\mathbb{Z}[i]$, the following properties of the norm function N hold for all $\alpha, \beta \in \mathbb{Z}[i]$:

1. $N(\alpha) \geq 0$.
2. $N(\alpha) = 0$ if and only if $\alpha = 0$.
3. $N(\alpha\beta) = N(\alpha)N(\beta)$.

Proof If we let $\alpha = a_1 + a_2 i$ and $\beta = b_1 + b_2 i$, these results are all straightforward computations. We leave the proof of these properties as an exercise (see Exercise 11). ◆

47.3 Lemma $\mathbb{Z}[i]$ is an integral domain.

Proof It is obvious that $\mathbb{Z}[i]$ is a commutative ring with unity. We show that there are no divisors of 0. Let $\alpha, \beta \in \mathbb{Z}[i]$. Using Lemma 47.2, if $\alpha\beta = 0$ then

$$N(\alpha)N(\beta) = N(\alpha\beta) = N(0) = 0.$$

Thus $\alpha\beta = 0$ implies that $N(\alpha) = 0$ or $N(\beta) = 0$. By Lemma 47.2 again, this implies that either $\alpha = 0$ or $\beta = 0$. Thus $\mathbb{Z}[i]$ has no divisors of 0, so $\mathbb{Z}[i]$ is an integral domain. ◆

Of course, since $\mathbb{Z}[i]$ is a subring of $\mathbb{C}$, where $\mathbb{C}$ is the field of complex numbers, it is really obvious that $\mathbb{Z}[i]$ has no 0 divisors. We gave the argument of Lemma 47.3 to

illustrate the use of the multiplicative property 3 of the norm function N and to avoid going outside of $\mathbb{Z}[i]$ in our argument.

47.4 Theorem The function ν given by $\nu(\alpha) = N(\alpha)$ for nonzero $\alpha \in \mathbb{Z}[i]$ is a Euclidean norm on $\mathbb{Z}[i]$. Thus $\mathbb{Z}[i]$ is a Euclidean domain.

Proof Note that for $\beta = b_1 + b_2 i \neq 0$, $N(b_1 + b_2 i) = {b_1}^2 + {b_2}^2$, so $N(\beta) \geq 1$. Then for all $\alpha, \beta \neq 0$ in $\mathbb{Z}[i]$, $N(\alpha) \leq N(\alpha)N(\beta) = N(\alpha\beta)$. This proves Condition 2 for a Euclidean norm in Definition 46.1.

It remains to prove the division algorithm, Condition 1, for N. Let $\alpha, \beta \in \mathbb{Z}[i]$, with $\alpha = a_1 + a_2 i$ and $\beta = b_1 + b_2 i$, where $\beta \neq 0$. We must find σ and ρ in $\mathbb{Z}[i]$ such that $\alpha = \beta\sigma + \rho$, where either $\rho = 0$ or $N(\rho) < N(\beta) = {b_1}^2 + {b_2}^2$. Let $\alpha/\beta = r + si$ for $r, s \in \mathbb{Q}$. Let q_1 and q_2 be integers in $\mathbb{Z}$ as close as possible to the rational numbers r and s, respectively. Let $\sigma = q_1 + q_2 i$ and $\rho = \alpha - \beta\sigma$. If $\rho = 0$, we are done. Otherwise, by construction of σ, we see that $|r - q_1| \leq \frac{1}{2}$ and $|s - q_2| \leq \frac{1}{2}$. Therefore

$$N\left(\frac{\alpha}{\beta} - \sigma\right) = N((r + si) - (q_1 + q_2 i))$$

$$= N((r - q_1) + (s - q_2)i) \leq \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2}.$$

Thus we obtain

$$N(\rho) = N(\alpha - \beta\sigma) = N\left(\beta\left(\frac{\alpha}{\beta} - \sigma\right)\right) = N(\beta)N\left(\frac{\alpha}{\beta} - \sigma\right) \leq N(\beta)\frac{1}{2},$$

so we do indeed have $N(\rho) < N(\beta)$ as desired. ◆

47.5 Example We can now apply all our results of Section 46 to $\mathbb{Z}[i]$. In particular, since $N(1) = 1$, the units of $\mathbb{Z}[i]$ are exactly the $\alpha = a_1 + a_2 i$ with $N(\alpha) = {a_1}^2 + {a_2}^2 = 1$. From the fact that a_1 and a_2 are integers, it follows that the only possibilities are $a_1 = \pm 1$ with $a_2 = 0$, or $a_1 = 0$ with $a_2 = \pm 1$. Thus the units of $\mathbb{Z}[i]$ are ± 1 and $\pm i$. One can also use the Euclidean Algorithm to compute a gcd of two nonzero elements. We leave such computations to the exercises. Finally, note that while 5 is an irreducible in $\mathbb{Z}$, 5 is no longer an irreducible in $\mathbb{Z}[i]$, for $5 = (1 + 2i)(1 - 2i)$, and neither $1 + 2i$ nor $1 - 2i$ is a unit. ▲

Multiplicative Norms

Let us point out again that for an integral domain D, *the arithmetic concepts of irreducibles and units are not affected in any way by a norm that may be defined on the domain.* However, as the preceding section and our work thus far in this section show, a suitably defined norm may be of help in determining the arithmetic structure of D. This is strikingly illustrated in *algebraic number theory*, where for a domain of *algebraic integers* we consider many different norms of the domain, each doing its part in helping to determine the arithmetic structure of the domain. In a domain of algebraic integers, we have essentially one norm for each irreducible (up to associates), and each such norm gives information concerning the behavior in the integral domain of the irreducible to

which it corresponds. This is an example of the importance of studying properties of elements in an algebraic structure by means of mappings associated with them.

Let us study integral domains that have a multiplicative norm satisfying Properties 2 and 3 of N on $\mathbb{Z}[i]$ given in Lemma 47.2.

47.6 Definition Let D be an integral domain. A **multiplicative norm** N **on** D is a function mapping D into the integers $\mathbb{Z}$ such that the following conditions are satisfied:

1. $N(\alpha) = 0$ if and only if $\alpha = 0$.
2. $N(\alpha\beta) = N(\alpha)N(\beta)$ for all $\alpha, \beta \in D$. ■

47.7 Theorem If D is an integral domain with a multiplicative norm N, then $N(1) = 1$ and $|N(u)| = 1$ for every unit u in D. If, furthermore, every α such that $|N(\alpha)| = 1$ is a unit in D, then an element π in D, with $|N(\pi)| = p$ for a prime $p \in \mathbb{Z}$, is an irreducible of D.

Proof Let D be an integral domain with a multiplicative norm N. Then

$$N(1) = N((1)(1)) = N(1)N(1)$$

shows that $N(1) = 1$. Also, if u is a unit in D, then

$$1 = N(1) = N(uu^{-1}) = N(u)N(u^{-1}).$$

Since $N(u)$ is an integer, this implies that $|N(u)| = 1$.

Now suppose that the units of D are *exactly* the elements of norm ± 1. Let $\pi \in D$ be such that $|N(\pi)| = p$, where p is a prime in $\mathbb{Z}$. Then if $\pi = \alpha\beta$, we have

$$p = |N(\pi)| = |N(\alpha)N(\beta)|,$$

so either $|N(\alpha)| = 1$ or $|N(\beta)| = 1$. By assumption, this means that either α or β is a unit of D. Thus π is an irreducible of D. ◆

47.8 Example On $\mathbb{Z}[i]$, the function N defined by $N(a + bi) = a^2 + b^2$ gives a multiplicative norm in the sense of our definition. We saw that the function ν given by $\nu(\alpha) = N(\alpha)$ for nonzero $\alpha \in \mathbb{Z}[i]$ is a Euclidean norm on $\mathbb{Z}[i]$, so the units are precisely the elements α of $\mathbb{Z}[i]$ with $N(\alpha) = N(1) = 1$. Thus the second part of Theorem 47.7 applies in $\mathbb{Z}[i]$. We saw in Example 47.5 that 5 is not an irreducible in $\mathbb{Z}[i]$, for $5 = (1 + 2i)(1 - 2i)$. Since $N(1 + 2i) = N(1 - 2i) = 1^2 + 2^2 = 5$ and 5 is a prime in $\mathbb{Z}$, we see from Theorem 47.7 that $1 + 2i$ and $1 - 2i$ are both irreducibles in $\mathbb{Z}[i]$.

As an application of mutiplicative norms, we shall now give another example of an integral domain that is *not* a UFD. We saw one example in Example 45.16. The following is the standard illustration.

47.9 Example Let $\mathbb{Z}[\sqrt{-5}] = \{a + ib\sqrt{5} \mid a, b \in \mathbb{Z}\}$. As a subset of the complex numbers closed under addition, subtraction, and multiplication, and containing 0 and 1, $\mathbb{Z}[\sqrt{-5}]$ is an integral domain. Define N on $\mathbb{Z}[\sqrt{-5}]$ by

$$N(a + b\sqrt{-5}) = a^2 + 5b^2.$$

(Here $\sqrt{-5} = i\sqrt{5}$.) Clearly, $N(\alpha) = 0$ if and only if $\alpha = a + b\sqrt{-5} = 0$. That $N(\alpha\beta) = N(\alpha)N(\beta)$ is a straightforward computation that we leave to the exercises (see Exercise 12). Let us find all candidates for units in $\mathbb{Z}[\sqrt{-5}]$ by finding all elements α in $\mathbb{Z}[\sqrt{-5}]$ with $N(\alpha) = 1$. If $\alpha = a + b\sqrt{-5}$, and $N(\alpha) = 1$, we must have $a^2 + 5b^2 = 1$ for *integers* a and b. This is only possible if $b = 0$ and $a = \pm 1$. Hence ± 1 are the only candidates for units. Since ± 1 are units, they are then precisely the units in $\mathbb{Z}[\sqrt{-5}]$.

Now in $\mathbb{Z}[\sqrt{-5}]$, we have $21 = (3)(7)$ and also

$$21 = (1 + 2\sqrt{-5})(1 - 2\sqrt{-5}).$$

If we can show that 3, 7, $1 + 2\sqrt{-5}$, and $1 - 2\sqrt{-5}$ are all irreducibles in $\mathbb{Z}[\sqrt{-5}]$, we will then know that $\mathbb{Z}[\sqrt{-5}]$ cannot be a UFD, since neither 3 nor 7 is $\pm(1 + 2\sqrt{-5})$.

Suppose that $3 = \alpha\beta$. Then

$$9 = N(3) = N(\alpha)N(\beta)$$

shows that we must have $N(\alpha) = 1$, 3, or 9. If $N(\alpha) = 1$, then α is a unit. If $\alpha = a + b\sqrt{-5}$, then $N(\alpha) = a^2 + 5b^2$, and for no choice of integers a and b is $N(\alpha) = 3$. If $N(\alpha) = 9$, then $N(\beta) = 1$, so β is a unit. Thus from $3 = \alpha\beta$, we can conclude that either α or β is a unit. Therefore, 3 is an irreducible in $\mathbb{Z}[\sqrt{-5}]$. A similar argument shows that 7 is also an irreducible in $\mathbb{Z}[\sqrt{-5}]$.

If $1 + 2\sqrt{-5} = \gamma\delta$, we have

$$21 = N(1 + 2\sqrt{-5}) = N(\gamma)N(\delta).$$

so $N(\gamma) = 1$, 3, 7, or 21. We have seen that there is no element of $\mathbb{Z}[\sqrt{-5}]$ of norm 3 or 7. This either $N(\gamma) = 1$, and γ is a unit, or $N(\gamma) = 21$, so $N(\delta) = 1$, and δ is a unit. Therefore, $1 + 2\sqrt{-5}$ is an irreducible in $\mathbb{Z}[\sqrt{-5}]$. A parallel argument shows that $1 - 2\sqrt{-5}$ is also an irreducible in $\mathbb{Z}[\sqrt{-5}]$.

In summary, we have shown that

$$\mathbb{Z}[\sqrt{-5}] = \{a + ib\sqrt{5} \mid a, b \in \mathbb{Z}\}$$

is an integral domain but not a UFD. In particular, there are two *different* factorizations

$$21 = 3 \cdot 7 = (1 + 2\sqrt{-5})(1 - 2\sqrt{-5})$$

of 21 into irreducibles. These irreducibles cannot be primes, for the property of a prime enables us to prove uniqueness of factorization (see the proof of Theorem 45.17). ▲

We conclude with a classical application, determining which primes p in $\mathbb{Z}$ are equal to a sum of squares of two integers in $\mathbb{Z}$. For example, $2 = 1^2 + 1^2$, $5 = 1^2 + 2^2$, and $13 = 2^2 + 3^2$ are sums of squares. Since we have now answered this question for the only even prime number, 2, we can restrict ourselves to odd primes.

47.10 Theorem **(Fermat's $p = a^2 + b^2$ Theorem)** Let p be an odd prime in $\mathbb{Z}$. Then $p = a^2 + b^2$ for integers a and b in $\mathbb{Z}$ if and only if $p \equiv 1 \pmod 4$.

Proof First, suppose that $p = a^2 + b^2$. Now a and b cannot both be even or both be odd since p is an odd number. If $a = 2r$ and $b = 2s + 1$, then $a^2 + b^2 = 4r^2 + 4(s^2 + s) + 1$, so $p \equiv 1 \pmod 4$. This takes care of one direction for this "if and only if" theorem.

For the other direction, we assume that $p \equiv 1 \pmod 4$. Now the multiplicative group of nonzero elements of the finite field $\mathbb{Z}_p$ is cyclic, and has order $p-1$. Since 4 is a divisor of $p-1$, we see that $\mathbb{Z}_p$ contains an element n of multiplicative order 4. It follows that n^2 has multiplicative order 2, so $n^2 = -1$ in $\mathbb{Z}_p$. Thus in $\mathbb{Z}$, we have $n^2 \equiv -1 \pmod p$, so p divides $n^2 + 1$ in $\mathbb{Z}$.

Viewing p and $n^2 + 1$ in $\mathbb{Z}[i]$, we see that p divides $n^2 + 1 = (n+i)(n-i)$. Suppose that p is irreducible in $\mathbb{Z}[i]$; then p would have to divide $n+i$ or $n-i$. If p divides $n+i$, then $n+i = p(a+bi)$ for some $a, b \in \mathbb{Z}$. Equating coefficients of i, we obtain $1 = pb$, which is impossible. Similarly, p divides $n-i$ would lead to an impossible equation $-1 = pb$. Thus our assumption that p is irreducible in $\mathbb{Z}[i]$ must be false.

Since p is not irreducible in $\mathbb{Z}[i]$, we have $p = (a+bi)(c+di)$ where neither $a+bi$ nor $c+di$ is a unit. Taking norms, we have $p^2 = (a^2+b^2)(c^2+d^2)$ where neither $a^2+b^2 = 1$ nor $c^2+d^2 = 1$. Consequently, we have $p = a^2+b^2$, which completes our proof. [Since $a^2+b^2 = (a+bi)(a-bi)$, we see that this is the factorization of p, that is, $c+di = a-bi$.] ◆

Exercise 10 asks you to determine which primes p in $\mathbb{Z}$ remain irreducible in $\mathbb{Z}[i]$.

EXERCISES 47

Computations

In Exercises 1 through 4, factor the Gaussian integer into a product of irreducibles in $\mathbb{Z}[i]$. [*Hint:* Since an irreducible factor of $\alpha \in \mathbb{Z}[i]$ must have norm >1 and dividing $N(\alpha)$, there are only a finite number of Gaussian integers $a+bi$ to consider as possible irreducible factors of a given α. Divide α by each of them in $\mathbb{C}$, and see for which ones the quotient is again in $\mathbb{Z}[i]$.]

1. 5 **2.** 7 **3.** $4+3i$ **4.** $6-7i$

5. Show that 6 does not factor uniquely (up to associates) into irreducibles in $\mathbb{Z}[\sqrt{-5}]$. Exhibit two different factorizations.

6. Consider $\alpha = 7+2i$ and $\beta = 3-4i$ in $\mathbb{Z}[i]$. Find σ and ρ in $\mathbb{Z}[i]$ such that

$$\alpha = \beta\sigma + \rho \qquad \text{with} \qquad N(\rho) < N(\beta).$$

[*Hint:* Use the construction in the proof of Theorem 47.4.

7. Use a Euclidean algorithm in $\mathbb{Z}[i]$ to find a gcd of $8+6i$ and $5-15i$ in $\mathbb{Z}[i]$. [*Hint:* Use the construction in the proof of Theorem 47.4.]

Concepts

8. Mark each of the following true or false.

_____ **a.** $\mathbb{Z}[i]$ is a PID.

_____ **b.** $\mathbb{Z}[i]$ is a Euclidean domain.

_____ **c.** Every integer in $\mathbb{Z}$ is a Gaussian integer.

_____ **d.** Every complex number is a Gaussian integer.

_____ **e.** A Euclidean algorithm holds in $\mathbb{Z}[i]$.

_____ **f.** A multiplicative norm on an integral domain is sometimes an aid in finding irreducibles of the domain.

_____ **g.** If N is a multiplicative norm on an integral domain D, then $|N(u)| = 1$ for every unit u of D.

_____ **h.** If F is a field, then the function N defined by $N(f(x)) =$ (degree of $f(x)$) is a multiplicative norm on $F[x]$.

_____ **i.** If F is a field, then the function defined by $N(f(x)) = 2^{(\text{degree of } f(x))}$ for $f(x) \neq 0$ and $N(0) = 0$ is a multiplicative norm on $F[x]$ according to our definition.

_____ **j.** $\mathbb{Z}[\sqrt{-5}]$ is an integral domain but not a UFD.

9. Let D be an integral domain with a multiplicative norm N such that $|N(\alpha)| = 1$ for $\alpha \in D$ if and only if α is a unit of D. Let π be such that $|N(\pi)|$ is minimal among all $|N(\beta)| > 1$ for $\beta \in D$. Show that π is an irreducible of D.

10. a. Show that 2 is equal to the product of a unit and the square of an irreducible in $\mathbb{Z}[i]$.

b. Show that an odd prime p in $\mathbb{Z}$ is irreducible in $\mathbb{Z}[i]$ if and only if $p \equiv 3 \pmod 4$. (Use Theorem 47.10.)

11. Prove Lemma 47.2.

12. Prove that N of Example 47.9 is multiplicative, that is, that $N(\alpha\beta) = N(\alpha)N(\beta)$ for $\alpha, \beta \in \mathbb{Z}[\sqrt{-5}]$.

13. Let D be an integral domain with a multiplicative norm N such that $|N(\alpha)| = 1$ for $\alpha \in D$ if and only if α is a unit of D. Show that every nonzero nonunit of D has a factorization into irreducibles in D.

14. Use a Euclidean algorithm in $\mathbb{Z}[i]$ to find a gcd of $16 + 7i$ and $10 - 5i$ in $\mathbb{Z}[i]$. [*Hint:* Use the construction in the proof of Theorem 47.4.]

15. Let $\langle\alpha\rangle$ be a nonzero principal ideal in $\mathbb{Z}[i]$.

a. Show that $\mathbb{Z}[i]/\langle\alpha\rangle$ is a finite ring. [*Hint:* Use the division algorithm.]

b. Show that if π is an irreducible of $\mathbb{Z}[i]$, then $\mathbb{Z}[i]/\langle\pi\rangle$ is a field.

c. Referring to part (b), find the order and characteristic of each of the following fields.

i. $\mathbb{Z}[i]/\langle 3\rangle$ **ii.** $\mathbb{Z}[i]/\langle 1 + i\rangle$ **iii.** $\mathbb{Z}[i]/\langle 1 + 2i\rangle$

16. Let $n \in \mathbb{Z}^+$ be square free, that is, not divisible by the square of any prime integer. Let $\mathbb{Z}[\sqrt{-n}] = \{a + ib\sqrt{n} \mid a, b \in \mathbb{Z}\}$.

a. Show that the norm N, defined by $N(\alpha) = a^2 + nb^2$ for $\alpha = a + ib\sqrt{n}$, is a multiplicative norm on $\mathbb{Z}[\sqrt{-n}]$.

b. Show that $N(\alpha) = 1$ for $\alpha \in \mathbb{Z}[\sqrt{-n}]$ if and only if α is a unit of $\mathbb{Z}[\sqrt{-n}]$.

c. Show that every nonzero $\alpha \in \mathbb{Z}[\sqrt{-n}]$ that is not a unit has a factorization into irreducibles in $\mathbb{Z}[\sqrt{-n}]$. [*Hint:* Use part (b).]

17. Repeat Exercise 16 for $\mathbb{Z}[\sqrt{n}] = \{a + b\sqrt{n} \mid a, b \in \mathbb{Z}\}$, with N defined by $N(\alpha) = a^2 - nb^2$ for $\alpha = a + b\sqrt{n}$ in $\mathbb{Z}[\sqrt{n}]$.

18. Show by a construction analogous to that given in the proof of Theorem 47.4 that the division algorithm holds in the integral domain $\mathbb{Z}[\sqrt{-2}]$ for $v(\alpha) = N(\alpha)$ for nonzero α in this domain (see Exercise 16). (Thus this domain is Euclidean. See Hardy and Wright [29] for a discussion of which domains $\mathbb{Z}[\sqrt{n}]$ and $\mathbb{Z}[\sqrt{-n}]$ are Euclidean.)